Visualizing the Cost Function: Examples

This section connects specific choices of parameters (`w`, `b`) to their resulting model (`f(x)`) and their corresponding cost (`J(w,b)`) on a contour plot.

The Core Relationship

- Every point on the contour plot represents a unique pair of parameters `(w, b)`.
- Each `(w, b)` pair defines a unique straight line, $f(x) = wx + b$.
- The position of the point on the contour plot tells you the cost (the "badness of fit") for that specific line. Points closer to the center have a lower cost.

Example Walkthrough

1. **Poor Fit (High Cost):**
    - A line with parameters far from the optimal values (e.g., a negative slope or an incorrect y-intercept) will not pass close to the data points.
    - The vertical distances (errors) between the line and the data points will be large.
    - Consequently, the calculated cost `J(w,b)` will be high. On the contour plot, this point will be on one of the outer ovals, far from the minimum.
2. **Good Fit (Low Cost):**
    - A line with parameters close to the optimal values will pass through the "middle" of the data points, minimizing the overall distance to them.
    - The vertical distances (errors) between the line and the data points will be small.
    - The calculated cost `J(w,b)` will be low. On the contour plot, this point will be very close to the center of the innermost oval, which represents the minimum possible cost.

The Next Step: Finding the Minimum Automatically

Manually guessing and checking parameters to find the minimum of the cost function is inefficient and impractical for complex models.

- **Goal:** We need an efficient algorithm that can automatically find the values of `w` and `b` that minimize the cost function $J(w, b)$.
- **Solution:** This algorithm is called **Gradient Descent**. It is one of the most fundamental and important algorithms in all of machine learning.